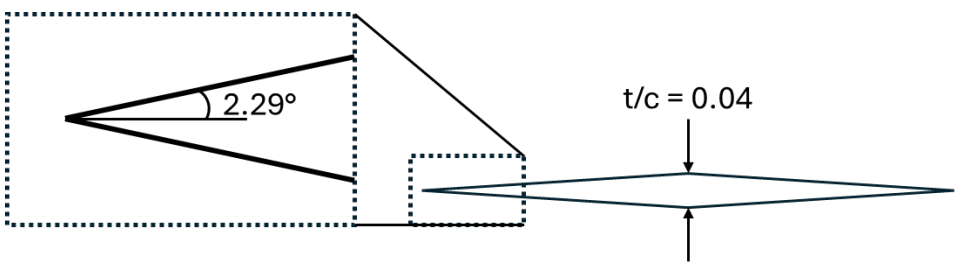
AE 462
Lab 6
Lab Session: Wednesday, March 25, 2026
Due: Wednesday, April 1, 2026, 9am CST

This laboratory assignment will focus on setting up solver configurations, mesh generation via Python, and simulating supersonic flows. The items required for your assignment submission are given here, alongside helpful instructions for completing this lab. As with previous assignments, all plots and presentations of data should include units, axis titles, and legends where appropriate. Provide your responses in a single PDF file. Upload any scripts, spreadsheets, or other work used during data analysis.

This laboratory exercise considers a diamond airfoil, having thickness of $t/c = 0.04$, in a M = 1.7 freestream flow. This airfoil configuration produces a surface half-angle oriented 2.29° relative to the chord line, as shown below.



The coordinates of the airfoil are given by the following relationship (+ for upper surface, - for lower surface):

$$\frac{y}{c} = \pm \begin{cases} 0.04\dfrac{x}{c} & 0 \le \dfrac{x}{c} < 0.5 \\ 0.04\left(1 - \dfrac{x}{c}\right) & 0.5 \le \dfrac{x}{c} \le 1 \end{cases}$$

1) **Comparison with Theory**
Shock-expansion theory is often used to analyze airfoils in supersonic flow. An approximate linear theory can also be developed assuming small angles and sufficiently weak Mach waves (which are both valid assumptions in this exercise). This linear theory can be distilled into expressions for the pressure coefficient (C_p), lift coefficient (C_l), and drag coefficient (C_d) for a 2D airfoil in inviscid flow (note α is given in radians):

$$C_p = \frac{2\theta}{\sqrt{M_\infty^2 - 1}}$$

$$C_l = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}}$$

$$C_d = \frac{4\alpha^2}{\sqrt{M_\infty^2 - 1}} + \frac{2}{\sqrt{M_\infty^2 - 1}}\left(\overline{\sigma_u^2} + \overline{\sigma_l^2}\right)$$

where

$$\theta_u = \frac{dy_u}{dx} - \alpha$$

$$\theta_l = -\frac{dy_l}{dx} + \alpha$$

$$\overline{\sigma_u^2} = \int_0^1 \left(\frac{dy_u}{dx}\right)^2 d\frac{x}{c}$$

$$\overline{\sigma_l^2} = \int_0^1 \left(\frac{dy_l}{dx}\right)^2 d\frac{x}{c}$$

For the case of M = 1.7 and α = 1 deg:

a. Co-plot the airfoil C_p distributions produced by linear theory and Euler (inviscid) CFD predictions. Be sure to label your axes, add a legend, and plot C_p on a reverse y axis (with negative values on top).
b. Provide the values of C_l and C_d produced by linear theory and Euler (inviscid) CFD predictions. State if any inconsistencies are observed between these two methods.

2) Aerodynamic Polar

Assuming the airfoil in inviscid flow and M = 1.7, across an angle-of-attack range:

a. Plot the C_l, C_d and C_m results obtained from Euler (inviscid) CFD predictions of the diamond airfoil across an angle of attack polar ranging from α = 0 deg to 10 deg, in 2 deg increments.
b. Calculate the lift-curve slope across the linear angle-of-attack region, with your answer given in C_l/rad. Compare this value to that produced by linear theory, from part 1) above.

3) Viscous Flow Analysis

Return to the baseline M = 1.7 and α = 1 deg case.

a. Assuming a RANS closure of the fully turbulent viscous flow, co-plot the viscous flow C_p distribution of the airfoil versus the inviscid (Euler) results obtained in problem 1) a.
b. Provide the values of C_l and C_d produced by viscous flow (RANS) and Euler (inviscid) CFD predictions. State if any inconsistencies are observed between these two methods.
c. You should observe a difference in C_d between the inviscid and viscous flow simulations. Provide a 1-2 sentence description as to why this difference may be observed.

Lab Instructions:

Organizing files

Create a new folder within your U:\AE 462 directory called "Lab 6". Download the files "diamond_template.cfg", "GMSH_diamond_unstructured.py", "diamond.su2", and "Visc_CFG_Additions.txt". Place these files in your U:\AE 462\Lab 6 directory.

Creating a Mesh

For this assignment, you will be using a Python script to generate your mesh. Modify the provided .py file to produce a mesh with 200 grid points on each edge of the diamond airfoil and 400 nodes total on the farfield (note that the variable num_nodes_farfield_arc corresponds to ¼ of the mesh circumference). Make sure the farfield radius is 80m.

Run this Python script. You should see a message indicating completion of the 1D mesh generation, after which point the Gmsh plugin will begin creating the unstructured surface mesh. After some time, a Gmsh window will be generated, and you will see your mesh in the graphical interface. Export your mesh as usual in .su2 format.

Configuring the Solver

For this lab, you will be running cases assuming both inviscid (Euler) and viscous (RANS) flow. Modify the diamond_template.cfg file with governing equations and boundary condition markers that are most appropriate for your problem. Items that you will need to input include:

- Physical governing equation (Euler, Navier-Stokes, RANS)
- Freestream Mach number
- Airfoil angle of attack
- Airfoil boundary condition marker
- Farfield boundary condition marker
- Mesh filename

First, configure the .cfg file for inviscid analysis, assuming the Euler equations. Save this .cfg file as "diamond_euler.cfg" after completing the necessary modifications. You will first run this case at $\alpha = 1$ deg to compare against aerodynamic theory (see problem 1)). Be sure to record your C_l and C_d values, as well as the surface C_p distribution. Next, you will run an angle of attack polar for this airfoil. Change the angle of attack and generate solutions for $\alpha = 0$, 2, 4, 6, 8, and 10 deg. For each angle of attack, record the C_l, C_d, and C_m values for later reporting. Note that each case should converge in ~300-400 iterations.

Finally, reconfigure the diamond_template.cfg file for viscous flow analysis, assuming the RANS equations. This time, use the provided "diamond.su2" mesh, as it is configured to appropriately resolve the boundary layer flow. In addition to changing the solver, boundary conditions, and mesh filename, you will also need to insert a few additional items in the .cfg file. Open the "Visc_CFG_Additions.txt" file and insert these items in the relevant sections. If any new sections are needed, place these before the "REFERENCE VALUE DEFINITION" section of the .cfg file. Now, run SU2 for the baseline case of M = 1.7 and $\alpha = 1$ deg. Take note that it will take a bit more time to converge than the inviscid case.

You now have the CFD-generated data needed to respond to the items required for the submission.